

— PRIVATE DATA · PRIVATE VERIFIER · ON-PREMISES

Qwen3.8 Agentic Search Evaluation on Private Data

Private data, private verifier, no data leaves the premises. Head to head against the model I actually use.

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THIS DECK



Qwen3.8 landed last week and my timeline has already decided it is better.

I have a default. For agentic search on my own hardware I reach for **Qwen3.6-35B-A3B**, and I have spent months tuning how it is served. A new model arrives, the hype arrives with it, and neither tells me whether to switch.

In agentic search, how much better is **Qwen3.8-27B** than **Qwen3.6-35B-A3B**, the model I currently like most and use by default?

the entire question this deck exists to answer

That framing matters. This is not a solo evaluation of a new model against a leaderboard. It is a replacement decision between two specific backends I have both already deployed, measured, and pushed to the wall of the same cheap GPU.

And I cannot answer it from a public benchmark, because the corpus I care about is **mine**.

THE TWO BACKENDS UNDER TEST

Both already pushed to the measured wall of one L4.

Qwen3.6-35B-A3B-MTP



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Revert README agent prompt block

d501a47 · 5 days ago

37 Commits

github/workflows	Publish prebuilt decode-optimized image to GHCR via Git...	2 months ago
docs	optimization-frontier: measured IQ4_XS speed AND qual...	2 months ago
patches	Fix audit findings: malformed patch headers + README ra...	2 months ago
scripts	Optional pre-baked tooling image: cold start ~3.5min -> ~...	2 months ago
Dockerfile	Reconcile tok/s numbers to one signal; refresh headline fr...	2 months ago
README.md	Revert README agent prompt block	5 days ago
docker-compose.yml	Default --ctx-size 56320 everywhere (ECC off is the assu...	2 months ago

README

Qwen3.6-35B-A3B-MTP on NVIDIA L4 24GB

(Q4_K_XL) + MTP speculative decoding on a single NVIDIA L4 24GB — GCP g2-standard-8 — via the official llama.cpp Docker image. Decode-optimized to ~91-99 tok/s (min ~91 chat, max ~99 math), lossless (full GPU residency + ECC-off).

gcp gguf gpu-inference inference-optimization large-language-models llama-cpp llm local-llm mixture-of-experts moe multi-token-prediction nvidia-l4 quantization qwen qwen3 speculative-decoding

Readme

sparse MoE · 35B total, ~3B active · 8 of 256 experts per token · UD-Q4_K_XL ~21.3 GiB resident · MTP head baked in

91-9956,320

tok/s decode, chat min to math maxtoken context

+45% over the 63 tok/s out-of-the-box config, lossless. Raw ceiling ~73 tok/s ECC-off; MTP multiplies it 1.27 to 1.37x. Two lossless wins: full GPU residency of all experts (+28%), ECC off (+10%).

WHY AN L4 AT ALL

24 GB, 72 W, ~300 GB/s, about \$0.24/hr spot on a g2-standard-8. Both repos are the result of pushing that one cheap card near its measured wall, and at 91 to 99 tok/s the 35B-A3B side is genuinely comfortable in a real agentic loop. ~92-100 tok/s at 56K against 23 tok/s at 104K is the throughput-versus-window trade the evaluation has to pay for. Measured 2026-08-14.

Qwen3.8-27B-UD-Q4_K_XL



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Serve on the VPC only, and add a restart path

d4f3984 · 16 hours ago

8 Commits

scripts	Serve on the VPC only, and add a restart path	16 hours ago
.gitignore	Qwen3.8-27B UD-Q4_K_XL on a single NVIDIA L4	2 days ago
LICENSE	Fix repo metadata: description, topics, licence, stale com...	2 days ago
README.md	Serve on the VPC only, and add a restart path	16 hours ago

README

Apache-2.0 license

Qwen3.8-27B · UD-Q4_K_XL · NVIDIA L4

Qwen3.8-27B serves at 23 tok/s decode over a 104,192-token context on a single NVIDIA L4 24 GB, using the Unsloth Dynamic v3.0 UD-Q4_K_XL GGUF and llama.cpp with MTP speculative decoding. That context is the largest the card will serve at this quantization.

Q4_K_XL GGUF) at 23 tok/s over a 104,192-token context on a single NVIDIA L4 24GB, llama.cpp + MTP speculative decoding. One-command GCP provisioning, tuning sweep, quality verification, and a binary-searched context ceiling.

benchmark gcp gguf llama-cpp llm-inference local-llm mtp nvidia-l4 quantization qwen qwen3 speculative-decoding

Readme

Apache-2.0 license

Activity

0 stars

dense-class, hybrid attention · 64 layers, full_attention_interval 4, so 16 hold a KV cache · native MTP head · positions to 262,144

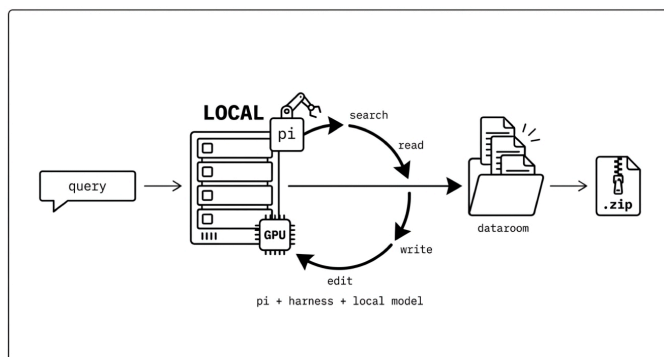
23104,192

tok/s decode token context, binary-searched ceiling

At 98% of the window, a 101,815-token prompt held 24.88 tok/s decode and 278 tok/s prefill, VRAM peak 24,082 MiB. Prefill 95 to 114 tok/s short, 278 at 102K. Bandwidth bound puts plain decode at ~16.8 tok/s; the rest is speculation.

Two systems, one to build the box and one to lock a model inside it.

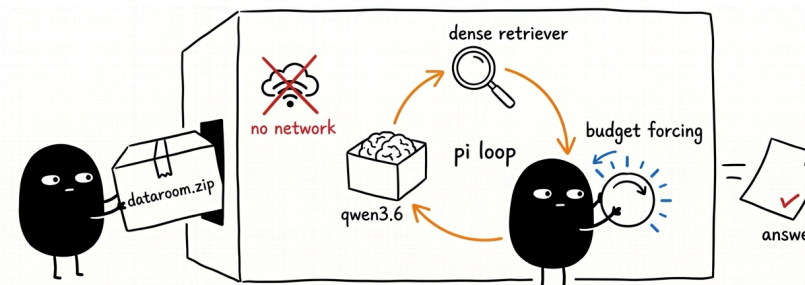
dataroom



A local model in a Pi harness loops search, read and write until it has built a comprehensive, fully cited dataroom on disk: a .zip you hand to the next model.

dataroom.hanxiao.io

searchbox



Locks a local Qwen in an **airgapped** loop with one .zip dataroom and no web, so it can only answer from what is in the box.

hanxiao.io/searchbox

One builds the private corpus. The other guarantees the model cannot cheat its way around it.

Six years of my own papers, plus everything the website says.

GROUP	FILES	DETAIL
papers	20	six years of published work, 2023-07 to 2026-07
model-cards	29	one spec sheet per released model
blog	161	tech 104 / press 28 / insights 14 / events 8 / kb 7
api	3	OpenAPI spec, usage guide, model selection guide
faq	2	163 question-answer pairs over 14 sources
site	2	27 page namespaces, 2,297 English strings
legal	1	terms, privacy, security, DPA

Snapshot 2026-08-13. 218 files total.

7,777

chunks indexed over 210 documents, median 91 units, longest 195 units at 2.1 KB

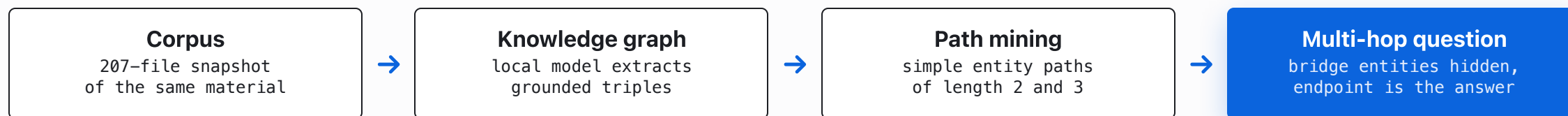
768

dimensions, jina-embeddings-v5-text-nano, L2-normalized

Papers and blog posts overlap by design: each major model has both a paper and a release blog, with different framing. That overlap is what makes multi-hop questions possible at all.

Two snapshots, stated separately. The ten audited graphs in the next section were built over an **earlier 207-file snapshot** of this same corpus. The 218-file snapshot above is what the agentic harness indexes today.

Questions mined out of the corpus, not written by hand.



A path of length k is a chain of k corpus-grounded facts, so path length is the difficulty knob. The question names the starting entity and hides the bridges, so a solver has to discover them rather than look one thing up.

207 documents in the snapshot these graphs were built over	10 graphs built over that corpus by the same local model	18,615 edges audited, every edge of every graph, no sampling	40 questions sampled per graph for the solver ladder
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Nobody vouches for a benchmark nobody wrote.

When a pipeline emits an evaluation set, acceptance rests on checks the pipeline can compute over its own artifact. The same three recur across systems:

COVERAGE

every source document contributes at least one item.

cheap, exact, model-free

DEDUPLICATION

no repeated items or edges in the artifact.

cheap, exact, model-free

GROUNDNESS

every claim traceable to a span of the corpus, in practice an exact substring check.

cheap, exact, model-free

A run that satisfies all three is reported as having produced a usable evaluation set. The one line of work that inspects acceptance closely, DoRA, abandons automation and sends items to three rounds of domain-expert review.

Before trusting a verifier to rank models, ask what licenses the claim that [the verifier itself is any good](#).

One word, two properties.

PROVENANCE

Is the evidence span an exact substring of the document it cites?
Checked with no model in the loop. This is what automatic pipelines mean by grounded.

a one-word span is trivially a substring

ASSERTION

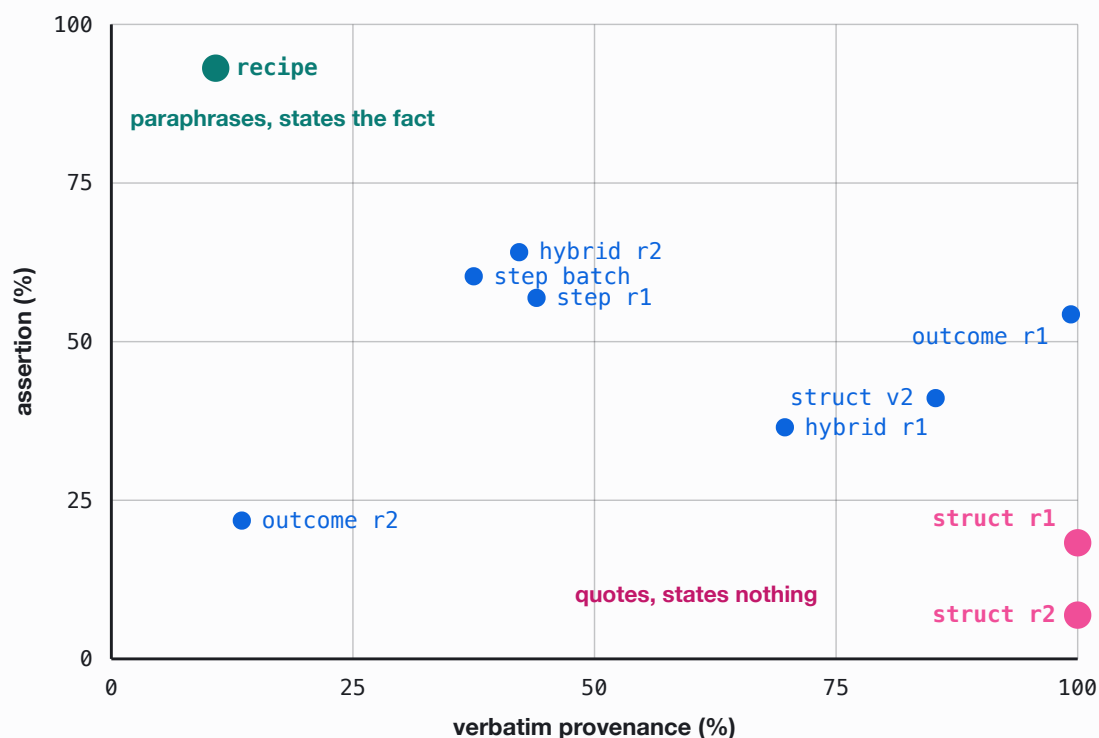
Shown the span alone, can a judge recover the subject, the relation and the object? The subject may be implicit; the relation and the object must be stated.

a keyword without the relation scores negative

Both are cheap. The substring check needs no model. The assertion audit needs one call of at most eight output tokens per edge, 2,811 calls for the largest graph, and all 18,615 edges of all ten graphs were audited at temperature 0 with no sampling.

Standard practice measures the first and reports it as the second.

The two axes invert at the extremes.



Struct r2 is **100% verbatim** and **6.9% asserting**.

median evidence span: 11 characters

Its builder set each evidence span to the object string itself.
Every edge passes the substring check while asserting nothing.

Recipe is **10.8% verbatim** and **93.1% asserting**.

last on provenance, first on assertion

Its spans paraphrase rather than quote, while stating the facts they support. Accepting on substring groundedness keeps the first graph and rejects this one.

A question is worth having when memory cannot answer it and the corpus can.

$$\text{usable}(q) = \neg \text{closed}(q) \wedge (\text{rag}(q) \vee \text{agentic}(q))$$

1

Not answerable closed-book.

Excludes items answerable from parametric memory, which measure recall rather than search.

2

Answerable from the corpus.

Excludes items that are broken, unanswerable, or whose gold answer is simply wrong.

Neither failure mode is visible to any artifact-level check. A graph's **usability** is the fraction of its sampled questions that are usable, and it is the quantity every acceptance metric has to be judged against.

Three conditions over one corpus.

CLOSED-BOOK	No retrieval at all. The model answers from what it already knows.	must fail
SINGLE-SHOT RETRIEVAL	BM25 top-5 passages placed in context, one shot, no follow-up query.	either may succeed
AGENTIC RETRIEVAL	The model issues its own BM25 queries, up to four rounds, before it answers.	either may succeed

Grading is string containment, falling back to an LLM equivalence judge. The ladder is applied identically to every graph and every question, with no graph-specific tuning, because it is the measuring instrument rather than a result.

The closed-book rung is the one that makes the set measure [search](#) instead of memory.

The most widely used check ranks the artifacts close to inversely.

ACCEPTANCE METRIC	PEARSON R	SPEARMAN P
Document coverage	−0.34	−0.42
Duplicate-free	−0.18	−0.22
Groundedness, verbatim	−0.74	−0.74
Edge count	+0.04	0.00
Groundedness, entailment	+0.05	+0.25
Capacity, log10 paths	+0.29	+0.15
Entity-entity share	−0.01	+0.01

Coverage, duplicate-freedom and edge count carry no usable information. Verbatim groundedness is the strongest correlate in the sample, with a negative sign: the three highest-provenance graphs are the three least usable.

The proposed substitutes are better behaved in sign and too weak to replace measurement. Capacity is dependable only at zero, where the one graph with no length-3 path yielded **one usable question in forty**.

n = 10 graphs. Ten points support ordinal statements, not precise coefficients.

A question set whose difficulty is a property of the corpus.

UNCONTAMINATED

Every item survived a closed-book probe on the same model family, so no item is answerable from parametric memory.

measures search, not recall

ANSWERABLE

Every item was reached by at least one retrieval-equipped solver, so a failure is the solver's rather than the question's.

a conservative lower bound

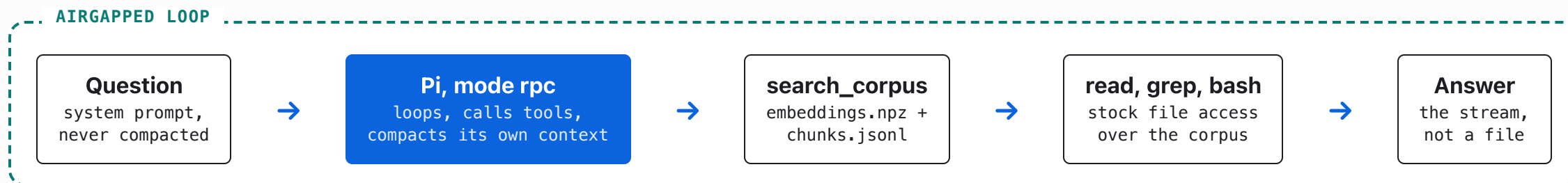
TUNABLE

Path length is the difficulty knob, and it is defined on graph structure rather than on question phrasing.

two hops and three hops

The same protocol reconstructs on any corpus. That is what makes it a verifier rather than a dataset: it is the procedure, applied to documents that never leave the building, and the 207-document corpus here is a shareable stand-in for the private one.

One agent, one external tool, no scaffold to credit.



WHAT THE HARNESS ADDS

A per-run binding of Pi to an OpenAI-compatible /v1, one extension registering `search_corpus`, and two wall-clock rails. No skills, no tool catalog, no supervision of what the model does with its turn.

WHAT IS SWITCHED OFF

`search_web` and `read_url` exist and are off by default, per run. With web access off the extension is never loaded and the model never sees the two tools.

A hit is a starting point, not an answer.

```
// one hit from search_corpus
{ score, path, lines, text }

// what the agent is told to do with it
read path around lines
widen the window while truncated
grep -n the same file for the topic
```

7777 x 768 matrix loaded in process: 117 ms to load, 4.3 ms per search.

The retrieval being tested is exactly one script, and everything else the agent does is stock file access. Nothing sits in between to credit or blame.

The instruction is load-bearing. The `lines` field alone was not enough: runs answered from the snippet without ever opening the file.

The extension is checked against the reference CLI on 15 query and filter combinations, including CJK: identical rankings, maximum score delta 1e-6.

Four times the throughput, or twice the window.

MEASURED ON ONE L4	3.6–35B–A3B	3.8–27B
Decode tok/s, by workload	92.3 to 100.4	22.9 to 24.6
Context, measured ceiling	56,320	104,192
Weights resident	~21.3 GiB	17.92 GB
Active params per token	~3B of 35B	dense-class 27B
KV-cache layers	all, heavy GQA	16 of 64
Speculation	MTP 1.27–1.37x	MTP, native head

Greedy, cache disabled, decode only. Measured 2026-08-14.

An agentic search loop spends its budget re-reading. Every round appends tool output and re-prefills a growing prefix, so the window bounds how many rounds fit before compaction starts discarding evidence, and decode speed bounds how long the whole thing takes.

The sparse model gives you **four times the tokens per second**. The dense-class model gives you **nearly twice the window** to spend them in.

The bandwidth arithmetic explains the floor, not the winner. 17.9 GB of weights at 300 GB/s bounds plain decode at about 16.8 tok/s on the 3.8; everything above that on either row is the MTP draft head, and speculative decoding is lossless by construction.

Which side of that trade wins on multi-hop retrieval is **an empirical question**, and it is the one this experiment asks.

Both ceilings are the card's, found by binary search and confirmed under load.

3.8 --CTX-SIZE	VRAM USED	FREE
65,536	21,500 MiB	3,070 MiB
90,112	23,138 MiB	1,432 MiB
98,304	23,684 MiB	886 MiB
102,400	23,956 MiB	614 MiB
104,192	24,076 MiB	494 MiB
104,448	fails to serve	

24,570 MiB device. Weights are 17.9 GB at every setting.

A context counts as working only when the server both reaches /health and completes a generation. llama.cpp allocates the KV cache at load time while some compute buffers are sized on the first decode, so a context that loads can still fail under traffic.

A 101,815-token prompt filling 98% of the window returned a correct answer at 24.88 tok/s decode and 278 tok/s prefill.
VRAM peaked at 24,082 MiB, container stayed healthy

The 35B-A3B ceiling is set the same way and lands at 56,320, first OOM at 57,344, also at ~98% VRAM. On both cards ECC off is the precondition that makes the top setting fit at all.

Two axes, held fixed everywhere else.

		CLOSED-BOOK	SINGLE-SHOT BM25	AGENTIC BM25, 4 ROUNDS
Qwen3.8-27B	104,192 ctx	control: excludes items answerable from memory	retrieval without a loop	the condition under test
Qwen3.6-35B-A3B	56,320 ctx	control, same corpus	retrieval without a loop	the incumbent to beat

HELD FIXED

The corpus, the question set, the mining and verbalization stage, the retrieval index, the grading procedure, and the four-round cap. Only the served model changes across rows.

SAMPLE

Per-graph samples are 40 questions, fewer where a graph is too sparse to fill the sample, so per-graph rates carry sampling error of several points.

Accuracy per turn, against two different budgets.

ANSWER SNAPSHOT	The answer the model holds at the end of every turn is captured, and carried forward on turns where it did not change.	per turn
JUDGE	Every turn's answer is graded against ground truth by the same local judge, not only the final one.	per turn
X-AXIS, TURNS	Accuracy against turn index. The legible unit: one turn is one round of the loop.	1 to budget
X-AXIS, TOKENS	Accuracy against cumulative fresh-prefill input tokens, summed compaction-safe from the session log.	the real budget

Turns are what a person reads; fresh-prefill input tokens are what the GPU pays. A model with a longer window can spend more tokens per turn, so the two axes can disagree about which model is more efficient, and both are reported.

The table exists. The cells do not.

MODEL, AGENTIC CONDITION	CLOSED-BOOK	SINGLE-SHOT	AGENTIC	TURNS TO ANSWER	FRESH-PREFILL TOKENS
Qwen3.8-27B-UD-Q4KXL-MTP					
Qwen3.6-35B-A3B-Q4KXL-MTP					
Difference	not yet measured				

 hatched cell = designed, not yet measured

Nothing on this machine contains a Qwen3.8 accuracy on this verifier. The rows, the columns and the metric definitions are fixed in advance so that the numbers cannot be chosen after the fact; the cells stay empty until the run happens.

● AWAITING MEASUREMENT

Three outcomes, three different conclusions.

QWEN3.8 AHEAD ON ACCURACY

The longer window is worth four times the decode cost on multi-hop retrieval, because more evidence survives to the answering turn.

then: window > throughput for this loop

35B-A3B AHEAD AT EQUAL WALL TIME

More turns beat a bigger window: the loop recovers by querying again, and compaction costs less than it appears to.

then: throughput > window for this loop

BOTH FLAT ABOVE TURN TWO

The verifier is not exercising the loop. Either the questions resolve in one retrieval or the four-round cap binds before the model needs it.

then: the instrument, not the models

The third outcome is the one worth stating out loud in advance, because it is a result about the verifier rather than about either model, and a result that only becomes visible if the per-turn curve is recorded rather than the final answer alone.

The instrument bounds the claim.

- | ONE CORPUS 207 documents from one organisation in one domain. Ordinal claims, not coefficients.
- | BM25 RETRIEVAL the solver ladder retrieves lexically, so a question unreached by any solver may still be answerable by a stronger system. Usability is a conservative lower bound.
- | FOUR ROUNDS the agentic condition is capped, and the cap is part of the measurement rather than a property of the model.
- | SAME-FAMILY JUDGE the judge comes from the same model family as the builders, which risks self-preference. The task is kept narrow and run at temperature 0.

One further constraint is already measured and worth carrying: across all ten construction runs, no builder ever invoked the local embedding or reranking tools available in its loop. Provided capability that is not directed goes unused, which is a warning about how much of an agentic result belongs to the agent.

**A benchmark you built yourself
is only worth what **a measurement
of its purpose** says it is.**

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THE HARNESS

